

Algorithmic Evaluation — Report

Data Exploration

The dataset is consistent with the assignment requirements and does not contain any issues with variable formats (`numeric`, `integer`, and `character`). Given its relatively small size, the data can be processed directly without additional optimization.

Two output columns are added to the dataset:

- `action` — the action assigned by the algorithm
- `reason` — a description of the rule(s) that led to the decision

Code Structure

For clarity and maintainability, the solution is structured using a modular, function-based approach. A central function processes the complete dataset and calls dedicated helper functions responsible for evaluating individual decision criteria.

Each helper function takes a single observation as input and returns the corresponding evaluation result, allowing the decision logic to remain clear and easy to follow.

Algorithm Logic

The dataset is evaluated row by row. For each observation, the decision criteria are tested in order of **action priority**, from highest to lowest.

Once an action is assigned, evaluation of that observation stops and the algorithm proceeds to the next row. This prevents lower-priority rules from overriding an already established decision.

If none of the preceding conditions are met, the observation is assigned the **STANDARD** action.

During the evaluation, the algorithm also records which criterion or criteria led to the final decision. A textual description of these conditions is stored in the `reason` variable, providing transparency into how each classification was determined.

Conclusion

The assignment is clearly defined and can be implemented as a deterministic rule-based classification algorithm. The explicit priority ordering of actions removes potential ambiguity when multiple conditions could apply to the same observation.